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Assessment of Numerical Forecasts for Hub-Height Wind Resource Parameters during an Episode of Significant Wind Speed Fluctuations

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Abstract: This study conducts a comprehensive evaluation of four scenario experiments using the CMA_WSP, WRF, and WRF_FITCH models to enhance forecasts of hub-height wind speeds at multiple wind farms in Northern China, particularly under significant wind speed fluctuations during high wind conditions. The experiments apply various wind speed calculation methods, including the Monin–Obukhov similarity theory (ST) and wind farm parameterization (WFP), within a 9 km resolution framework. Data from four geographically distinct stations were analyzed to assess their forecast accuracy over a 72 h period, focusing on the transitional wind events characterized by substantial fluctuations. The CMA_WSP model with the ST method (CMOST) achieved the highest scores across the evaluation metrics. Meanwhile, the WRF_FITCH model with the WFP method (FETA) demonstrated superior performance to the other WRF models, achieving the lowest RMSE and a greater stability. Nevertheless, all models encountered difficulties in predicting the exact timing of extreme wind events. This study also explores the effects of these methods on the wind power density (WPD) distribution, emphasizing the boundary layer’s influence at the hub-height of 85 m. This influence leads to significant variations in the central and coastal regions. In contrast to other methods that account for the comprehensive effects of the entire boundary layer, the ST method primarily relies on the near-surface 10 m wind speed to calculate the hub-height wind speed. These findings provide important insights for enhancing wind speed and WPD forecasts under transitional weather conditions.

Keywords: wind energy; mesoscale models; boundary layer; fluctuation; model evaluation



Citation: Mo, J.; Shen, Y.; Yuan, B.; Li, M.; Ding, C.; Jia, B.; Ye, D.; Wang, D. Assessment of Numerical Forecasts for Hub-Height Wind Resource Parameters during an Episode of Significant Wind Speed Fluctuations. *Atmosphere* **2024**, *15*, 1112. <https://doi.org/10.3390/atmos15091112>

Academic Editors: Paulo Rocha, Bahram Gharabaghi and Massimiliano Burlando

Received: 3 July 2024

Revised: 1 September 2024

Accepted: 11 September 2024

Published: 13 September 2024



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1. Introduction

As the global demand for renewable energy accelerates, wind power, a critical component of clean energy, has seen rapid development in China. This growth, driven by the “dual carbon” objectives, led to China achieving a world-leading installed capacity of 370 million kilowatts in 2022, marking an 11.2% increase compared to the previous year [1]. The extensive construction of wind farms significantly contributes to energy conservation and emission reduction, while altering the surface roughness and aerodynamics of the area of construction [2]. These alterations impact land surface processes, near-surface flux exchange, and boundary layer characteristics, and result in pronounced wake and edge effects [3]. Such effects have substantial ramifications on the meteorological and atmospheric environmental conditions, including the wind speed, turbulence, temperature, and humidity [4]. Wind energy, now a mainstream resource due to decades of research and technological advancement, still requires continued innovation to meet global clean

energy demands, particularly through enhancing our understanding of atmospheric flow, through the optimization of large-scale wind machines [5]. Moreover, the need for accurate wind speed and energy predictions is increasingly pivotal in sectors like electric grid management, power trading, and energy security [6].

Accurately predicting the hub-height wind speed is crucial for optimizing wind turbine efficiency and ensuring power grid safety [7,8]. However, the inherent variability and randomness of wind speeds pose significant challenges in numerical weather prediction (NWP). Current practices for wind power forecasts exceeding 24 h involve a two-stage process: first, predicting wind speeds using numerical weather prediction (NWP) models, and second, converting these wind speeds into power-generation estimates. The accuracy of the wind speed predictions contributes approximately 70% to the overall accuracy of wind power forecasts [9].

The heights of model layers are varying and different from the typical turbine hub-height. Thus, traditional methods for estimating vertical wind speeds, such as using the nearest-height model layer, interpolating between different model layers, or applying the power-law exponent method, face limitations under complex terrain and varying atmospheric stability conditions [10–12]. Despite many studies estimating hub-height wind speeds and predicting wind power density based on surface wind speeds, significant uncertainties remain [13,14]. Moreover, the increasing frequency of extreme weather events spurred by global warming, such as high-temperature heat waves and extreme winter heating demand, poses additional challenges to power systems, especially in managing the variability of wind power output [15,16]. Therefore, the accurate prediction of wind speeds at hub-height is crucial for investigating the aforementioned challenges.

The Monin–Obukhov similarity theory is currently the principal method in engineering for calculating hub-height wind speeds. As a foundational concept in boundary layer meteorology, this theory is essential to the near-surface layer parameterization process of NWP models. Utilizing this theoretical framework, researchers have accounted for variations in surface roughness within the constant flux layer to more accurately calculate vertical wind profiles for wind farms [17,18]. Bahamonde and Litrán considered variations in atmospheric stability, friction velocity, and aerodynamic roughness to extrapolate hub-height wind speeds to an 80 m height from near-surface speeds, providing a detailed analysis of wind speed deviations under different atmospheric conditions [19]. Additionally, Barthelmie explored the influence of surface roughness changes due to tides on offshore wind farms' vertical wind profiles, identifying an uncertainty in offshore wind energy resource assessment at a 50 m height of approximately 8% [20,21]. While many studies derive hub-height wind speeds from near-surface data during post-processing, few directly compute these results online in numerical models, highlighting that integrating advanced similarity theory, including roughness and stability factors, into NWP models to enhance wind speed forecast accuracy is a promising area for further research.

In addition to implicit representations, numerical models also incorporate explicit parameterization schemes for wind farms. A notable example is the Fitch WFP scheme, which is integrated within the boundary layer parameterization of the weather research and forecasting (WRF) model, which characterizes wind farms through the combination of momentum sinks and turbulent kinetic energy sources. This scheme has been evaluated and refined by numerous scholars [22–27]. Its online coupling approach facilitates the direct feedback of changes in corresponding meteorological elements to other physical processes, enhancing the NWP model's capacity to accurately simulate the hub-height wind speed and turbulence effects around wind farms. Olson improved the wind farm and related physical parameterization schemes in the second wind forecast improvement project (WFIP2) in the northwest of the United States, validating the NWP model's 0–24 h forecasts with 80 m SODAR data from 19 sites, and noted that the enhancements reduced the mean absolute error by 4–30% [8]. Lee et al., utilizing the Fitch scheme at a Central American wind farm, found that their forecasts closely aligned with the actual wind speed and turbulence trends, substantially reducing the power-prediction errors during periods of rapid wind

speed fluctuations [27]. Other explicit WFP schemes include the wake parameterization (EWP) and the generalized actuator disk models [28]. Larsén and Fischereit evaluated an EWP method in the WRF model using aircraft and Fino1 observation data, recording wind speed reductions within a wind farm that were between 0.5 to 2 m/s and noting an underestimation of the wind speed and TKE [29].

However, the current NWP studies predominantly focus on the meteorological and climatic effects of wind farms [3], with few examining the accuracy of hub-height wind speed forecasts or evaluating different prediction methods during the same period [11,27]. Research on the characteristics of wind speed fluctuations, particularly the turning points of wind increases and decreases, is also rare. Additionally, the existing research results include a degree of uncertainty due to the scarcity of vertical wind speed observations and short time series, as well as the significant variability introduced by local topography.

In the context of global warming, particularly in recent years, hot summers have led to a significant increase in electricity consumption for air conditioning, frequently setting new records for summer power loads. Consequently, the variability of wind power poses a substantial challenge to grids' ability to handle peak demands, while wind transitional weather can cause additional scheduling challenges [30,31]. In light of this, based on observations from four wind farms in northern China, our study focuses on a peak period of summer electricity use from July 10 to 13, 2022, during a significant wind speed fluctuation period. The aim is to investigate two hub-height wind speed approaches and comprehensively evaluate their performance in terms of accuracy for hub-height wind speed and wind energy resource forecasting.

Our goal is to provide the energy industry with a more precise data source for hub-height wind speed prediction, thus enhancing the operational efficiency of wind farms and the stability of the power grid, while also laying a scientific foundation for effective wind farm management and the optimized utilization of wind energy resources. This paper is structured as follows: an introduction to the NWP models and the theoretical basis of the ST and WFP methods; details of the wind farm observation data; a presentation of the research results and discussions, which include a comparison of the two methods for the statistical indices, wind speed fluctuation processes, the spatial distributions of wind power density, and the influence of the boundary layer; and a summary of the key findings.

2. Models and Methods

2.1. M-O Similarity Theory (ST) Method

Since its establishment in 1954, the M-O theory has been extensively applied across various disciplines for both theoretical studies and engineering applications [32]. The M-O theory assumes that the layer near the Earth's surface functions as a quasi-constant flux layer, typically integrated within the NWP models [17]. The basic principles of similarity theory are already included in the near-surface-layer parameterization schemes of the WRF model for the calculation of 10 m wind speed. We have extracted relevant variables such as the Monin–Obukhov (M-O) length, surface roughness, and incorporated atmospheric stability adjustments. Through a collaborative effort between the national Public Meteorological Service Center (PMSC) and the Beijing Institute of Urban Meteorology (IUM), these modifications were integrated into the Revised MM5 near-surface-layer parameterization scheme [33], enabling the model to directly compute hub-height wind speeds and produce an output for wind speeds at 50 m, 70 m, and 100 m. Detailed formulas are provided below.

Under the basic of similarity theory, formula Equation (1) computes the wind speed U_h at a height h above the ground in a neutral atmospheric layer as:

$$u_h = \frac{U_*}{k} \times \ln\left(\frac{h}{z_0}\right) \quad (1)$$

where k represents von Kármán's constant, and z_0 denotes the surface roughness. Due to technical limitations in the direct coupling method within the WRF model, this study utilizes the default surface roughness derived from land processes. Adjustments for wind

farms' equivalent roughness were not considered. U_* represents the friction velocity at the surface. Additionally, the vertical change in wind speed is influenced by atmospheric stratification, leading to the inclusion of an atmospheric stability adjustment term $\Psi(\tau)$ in Equation (2) [17]:

$$u_h = \frac{U_*}{k} \times \left(\ln\left(\frac{h}{z_0}\right) - \Psi(\tau) \right) \quad (2)$$

where $\Psi(\tau)$ is a function of the stability parameter τ , computed based on a universal function form established from extensive field observations [34].

For unstable atmospheric conditions ($\tau < 0$), Equation (3) is used to calculate $\Psi(\tau)$:

$$\begin{cases} \Psi(\tau) = 2\ln\left(\frac{1+x}{2}\right) + \ln\left(\frac{1+x^2}{2}\right) - 2\arctan(x) + \frac{\pi}{2} \\ x = (1 - 15\tau)^{1/4} \end{cases} \quad (3)$$

For stable atmospheric conditions ($\tau \geq 0$), Equation (4) is utilized:

$$\Psi(\tau) = -4.7\tau \quad (4)$$

where $\tau = h/L$ and L represents the Monin–Obukhov length, which is directly obtained from the near-surface-layer parameterization scheme in the WRF model. By implementing Equations (2)–(4), we can calculate wind speeds at various heights and establish the ratio between near-surface and hub-height wind speeds using Equation (5).

$$\frac{U_2}{U_1} = \frac{\ln\left(\frac{h_2}{z_0}\right) - \Psi(\tau)}{\ln\left(\frac{h_1}{z_0}\right) - \Psi(\tau)} \quad (5)$$

To streamline the terminology, the following paragraphs collectively abbreviate the Monin–Obukhov similarity theory as the “ST method”.

2.2. Wind Farm Parameterization (WFP) Method

The wind farm parameterization discussed in this paper represents the Fitch wind turbine drag parameterization scheme, which has been successfully integrated into the official release of the WRF numerical model for online coupling [23,35,36]. For conciseness in terminology, this study refers to the wind farm parameterization method as the “WFP method”.

The WRF model activates the Fitch scheme via namelist settings, specifying crucial turbine parameters such as the hub-height, blade diameter, cut-in and cut-out wind speeds, and power curve in the `windturbine.txt` and `wind-turbine-1.tbl` files. For general parameter settings of the wind turbines, readers can refer to the related article [4]. The specific parameters used for simulation at the four wind farms are detailed in the next section. To address the issue of excessive turbulence observed during the validation of the Fitch scheme, this paper adopts an empirical turbulence constraint factor for wind farms, globally adjusting C_{TKE} to 25% of its original value [25], which enables the application of the turbulence advection scheme.

Originating from the MYNN boundary layer and turbulence scheme [37], the Fitch scheme facilitates the conversion of diminished kinetic energy (KE) into electrical energy (p) and turbulence kinetic energy (TKE). This transformation process relies on calculating the wind turbine thrust coefficient C_T and power coefficient C_P , which both depend on the wind speed and the specific type and specifications of the turbine.

Consequently, the turbulence-conversion coefficient can be derived as follows:

$$C_{TKE} = C_T - C_P \quad (6)$$

The resistance generated by the turbine within the boundary layer flow can be represented by the drag equation:

$$F_{drag} = \frac{1}{2} C_T (|V|) \rho |V| V A \quad (7)$$

Here, $V = (u, v)$ represents the horizontal velocity vector, ρ is the air density, and $A = (\pi/4) \times D^2$ is the cross-sectional area of the rotor, with D denoting the diameter of the turbine blades.

The kinetic energy loss within a single grid point of the numerical model can be expressed as:

$$\frac{\partial KE_{drag}^{ijk}}{\partial t} = -\frac{1}{2} N_t^{ij} \Delta x \Delta y C_T (|V|_{ijk}) \rho |V|_{ijk}^3 A_{ijk} \quad (8)$$

Here, N_t represent the turbine numbers; i, j , and k denote the grid indices in the three-dimensional model space; Δx and Δy represent the horizontal grid spacing; and A_{ijk} refers to the cross-sectional area swept by the wind turbine blades between vertical layers k and $k + 1$ at the grid (i, j) .

Thus, the rate of change for the grid cell's kinetic energy loss is:

$$\frac{\partial KE_{cell}^{ijk}}{\partial t} = \frac{\partial}{\partial t} \frac{\rho_{ijk} |V|_{ijk}^2}{2} (z_{k+1} - z_k) \Delta x \Delta y \quad (9)$$

Furthermore, the momentum loss rate is defined as:

$$\frac{\partial |V|_{ijk}}{\partial t} = \frac{\frac{1}{2} N_t^{ij} C_T (|V|_{ijk}) |V|_{ijk}^2 A_{ijk}}{(z_{k+1} - z_k)} \quad (10)$$

The rates of change for the converted electrical energy (p) and turbulence kinetic energy (TKE) are determined as follows:

$$\frac{\partial P_{ijk}}{\partial t} = \frac{\frac{1}{2} N_t^{ij} C_P (|V|_{ijk}) |V|_{ijk}^3 A_{ijk}}{z_{k+1} - z_k} \quad (11)$$

$$\frac{\partial TKE_{ijk}}{\partial t} = \frac{\frac{1}{2} N_t^{ij} C_{TKE} (|V|_{ijk}) |V|_{ijk}^3 A_{ijk}}{(z_{k+1} - z_k)} \quad (12)$$

2.3. NWP Models and Scenarios Setup

The China Meteorological Administration wind and solar professional model (CMA_WSP) has been developed under the framework of the WRF model (version 4.2.2), incorporating specific enhancements to the parameterization schemes for wind and solar energy forecasting. The production of 70, 80, 100, and 120 m-height wind speed data by the CMA_WSP is conducted using the advanced hub-height wind speed method previously mentioned. As an operational system, the CMA_WSP MODEL supports the CMA's dedicated wind and solar professional meteorological service, incorporating specific enhancements tailored for the energy sector. The simulation area of the latest generation of the CMA_WSP MODEL covers the whole of China (Figure 1a), providing a resolution of 9 km and predicting 336 h ahead with 15 min intervals. In this paper, the wind speed heights are changed to 65, 80, and 90 m.

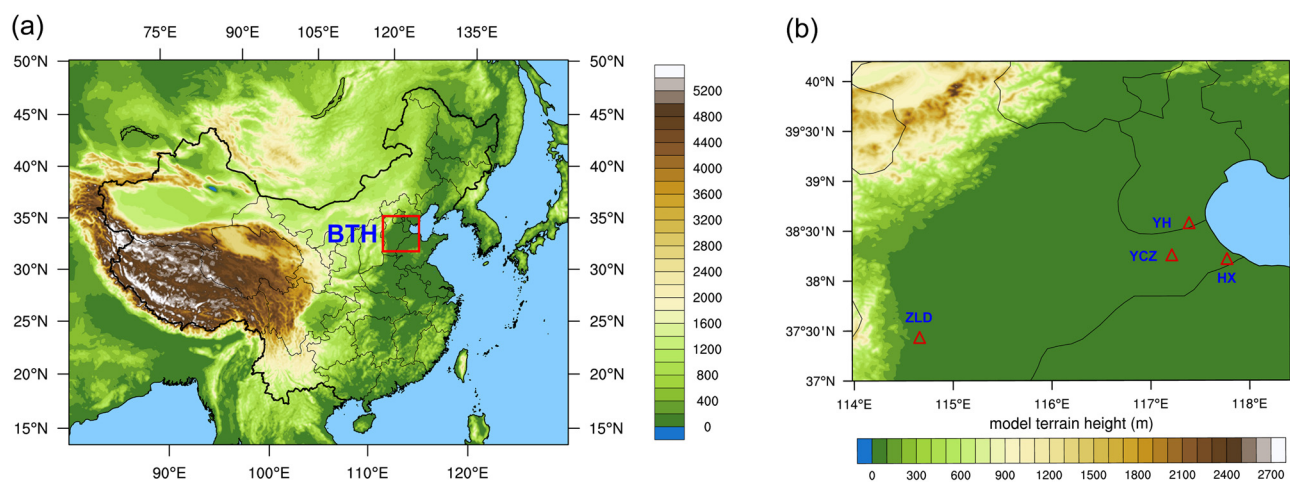


Figure 1. (a) Model domain and BTH region in red rectangle (b) Zoomed-in BTH region and wind farm locations (red triangles) with terrain altitude.

The domain setup of WRF/WRF_FITCH (version 4.2.2) is the same as that of the CMA_WSP MODEL, and the research area focuses on the Beijing–Tianjin–Hebei region (BTH). A zoomed-in view of the BTH region with wind farm positions is shown in Figure 1b. The model features enhanced vertical resolution near the surface with intervals of 20 m, including 21 layers below 1 km, 9 layers below 200 m, and 5 layers within 100 m. All models’ boundary and initial conditions are obtained from the European Center for Medium-Range Forecasts (ECMWF). The third-generation reanalysis dataset, ERA5, offers improved spatial and temporal resolutions compared to its predecessor, ERA-Interim. It features a horizontal resolution of 0.28125°, 137 levels in the vertical direction (with the highest resolution reaching 0.01 hPa), and an hourly temporal resolution. The data are processed using the 4DVar assimilation technique of the IFS cycle [38]. ERA5 is also recommended by the WMO GCOS (World Meteorological Organization Global Climate Observing System) and is included in the data sources for upper-air wind speed and direction. In order to facilitate comparison with the CMA_WSP MODEL, the parameterization schemes of the WRF models are aligned with those of the CMA_WSP MODEL. Detailed physical parameterization schemes can be found in Table 1.

Table 1. Main physical parameterization schemes.

Parameterization Scheme	Option
Microphysics Parameterization	Thompson [39]
Convection Parameterization	New Tiedtke [40]
Long/Shortwave Radiation	RRTMG [41]
Near-Surface	Revised MM5 [33]
Land Surface	Noah [42]

In this study, four scenarios were designed based on WRF 4.2.2 and the CMA_WSP MODEL to ensure a fair comparison (Table 2). Given that the CMA_WSP MODEL has undergone extensive enhancements tailored to energy meteorology on the WRF 4.2.2, its results were included as a secondary dataset for comparison. The other three scenarios, all utilizing WRF 4.2.2, were treated as a comparable series. Hub-height wind speeds in the CMA_WSP model were obtained using the ST method, referred to as CMOST. For the WRF 4.2.2 model without the Fitch scheme, the ST method was used to calculate hub-height wind speeds, and is referred to as WMOST. The WRF version 4.2.2 model with the Fitch scheme enabled that also used eta-level interpolation for hub-height wind speeds is referred to as FETA. Finally, the WRF version 4.2.2 model without the Fitch scheme and for

which eta-level interpolation was applied to derive hub-height wind speeds is referred to as WETA.

Table 2. Scenario configuration and hub-height wind speed calculation method.

Scenario	Model	WFP	Hub-Heights Wind
CMOST	CMA_WSP	/	ST based on 10 m
WMOST	WRF	OFF	ST based on 10 m
FETA	WRF_FITCH	ON	Interpolation from eta levels
WETA	WRF	OFF	Interpolation from eta levels

2.4. Wind Farm Observation Data

The wind speed observation data at the height of the wind turbine hubs and information on the models of wind turbines within the wind farms were obtained from power stations belonging to the South Hebei Power Grid Company. Due to security and confidentiality reasons, only a brief introduction is provided regarding those limitations. The four wind farms, abbreviated as YH, YCZ, HX, and ZLD, are located in the southwestern mountainous area and the eastern coastal region of Hebei Province. The hub-heights, which also correspond to the heights observed by the anemometer towers, are 90 m for YH and ZLD, 65 m for HX, and 80 m for YCZ. Figure 1b illustrates their specific locations. ZLD is situated in the plains at the eastern foot of the Taihang Mountains, while the other three wind farms are located in the eastern coastal area. The measured wind speed data for these wind farms are collected from meteorological towers surrounding each farm at 15 min intervals. In this study, wind speed data at the hub-heights are used to validate and analyze model predictions. When validating the hub-height results for different wind farms, the outputs from all scenarios were standardized to 65, 80, and 90 m using the hub-height wind calculation methods outlined in Table 2.

3. Results and Discussions

3.1. Comparative Forecast Evaluation of Different Scenario

Our study focuses on a peak period of summer electricity use, from 9 to 12 July 2022, to investigate two wind speed forecasting methods at different hub-heights. During this period, the examined wind farms experienced significant wind speed fluctuations, ranging from low (less than 3 m/s) to high (more than 8 m/s), with the maximum wind speed reaching 16 m/s, and displaying significant variability. Both models were initiated at 12:00 UTC on 9 July 2022, with a 4 h spin-up period, and their forecasting efficacy for the subsequent 72 h, from 16:00 on 9 July to 16:00 on 12 July 2022, was evaluated.

The verification of the four scenarios involved standard statistical indicators commonly employed in numerical models, such as the root mean square error (RMSE) and correlation coefficient (R). Additionally, performance evaluation indicators from the Energy Regulatory Bureau of National Energy Administration, including the accuracy rate (AR) and qualification rate (QR), were incorporated [43]. The RMSE indicates the magnitude of the forecasting error for the wind speed, the R signifies the degree of correlation between the forecasted and actual measured values, the AR represents the closeness between consecutive wind speed forecast values and the actual measurements over a specified period, and the QR measures the proportion of wind speed forecasts that meet the basic evaluation criteria within a given time frame. The mathematical expressions for these indicators are as follows:

$$RMSE = \left(\sqrt{\frac{1}{n} \sum_{i=1}^n (V_f^i - V_o^i)^2} \right) \quad (13)$$

$$R = \frac{\sum_{i=1}^n (V_f^i - \bar{V}_f)(V_o^i - \bar{V}_o)}{\sqrt{\sum_{i=1}^n (V_f^i - \bar{V}_f)^2} \sqrt{\sum_{i=1}^n (V_o^i - \bar{V}_o)^2}} \quad (14)$$

$$A = \left(1 - \sqrt{\frac{1}{n} \sum_{i=1}^n \left(\frac{V_f^i - V_o^i}{V} \right)^2} \right) \times 100 \quad (15)$$

$$Q = \frac{1}{n} \sum_{i=1}^n Q_i \times 100\% \quad (16)$$

$$Q_i = \begin{cases} 0, & V_f^i - V_o^i \geq 0.3 \\ 1, & V_f^i - V_o^i < 0.3 \end{cases} \quad (17)$$

Figure 2 presents the validation results for the averaged forecasts and observed data across four wind farms in the BTH region over the initial 0–72 h, focusing on the collective error of the overall data. In this paper, all references to average data refer to this calculation. These averages reflect the predictive accuracy of different scenario experiments across the entire region. Meanwhile, the individual performance of each station is presented separately in Table 3.

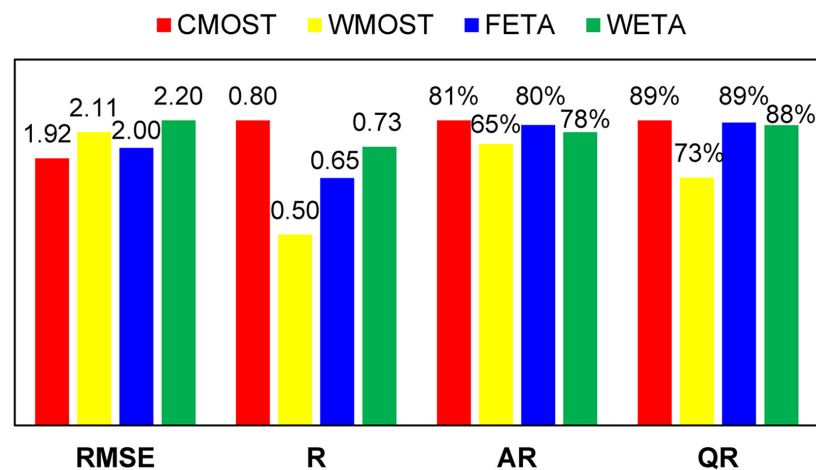


Figure 2. The averaged statistical indicators of the BTH region for the 0–72 h forecast period.

Table 3. Single-station verification results. The values with the lowest relative deviation in the same station are highlighted in bold.

INDEX	STATION	CMOST	WMOST	FETA	WETA
RMSE	ZLD	2.56	2.03	1.89	2.64
	YCZ	2.75	3.56	3.68	3.26
	HX	4.65	4.16	4.66	4.68
	YH	3.31	4.19	3.71	3.43
R	ZLD	0.75	0.71	0.81	0.82
	YCZ	0.54	0.11	0.15	0.27
	HX	0.65	0.53	0.60	0.66
	YH	0.46	0.12	0.28	0.33
AR	ZLD	74%	80%	81%	74%
	YCZ	73%	69%	67%	70%
	HX	53%	58%	54%	53%
	YH	73%	67%	70%	72%
QR	ZLD	80%	69%	90%	73%
	YCZ	74%	54%	65%	66%
	HX	31%	36%	30%	30%
	YH	73%	60%	72%	77%

For the region-wide averages, the CMOST method achieved the best performance of RMSE indicator among the scenarios with an average value of 1.92 m/s, followed closely by FETA at 2.0 m/s, WMOST at 2.11 m/s, and WETA at 2.20 m/s. In terms of the R indicator, CMOST also demonstrated a relatively high correlation coefficient of 0.8, whereas WMOST had a lower correlation of only 0.5. Although the same hub-height wind speed calculation method was used, the significant difference between CMOST and WMOST in correlation coefficients might be attributed to other improved physical processes, such as radiation schemes, within the CMA_WSP model. The other two WRF-based experiments both achieved correlation coefficients exceeding 0.60. For the AR and QR indicators, CMOST and FETA showed higher accuracy, with both performing similarly well, while WMOST and WETA had slightly lower accuracy.

For individual sites, as shown in Table 3, the performance of different scenario experiments varies across the various bias indicators, with each experiment showing its strengths in specific cases. Regarding the RMSE indicator, FETA performs best at the ZLD and YH wind farms, while CMOST excels at the YCZ wind farm, and WMOST at the HX wind farm. In terms of the R indicator, WETA shows the best performance at the ZLD and HX wind farms, whereas CMOST outperforms the others at the YCZ and YH wind farms. Overall, the correlation coefficients are higher at the ZLD and HX sites, and lower at the YCZ and YH wind farms. Concerning the AR and QR indicators, where the grid dispatch minimum requirement is 60% [44], all four experiments exceed 70% at the ZLD, YCZ, and YH sites, with the FETA method achieving a QR of 90% at the ZLD wind farm. However, at the HX wind farm, all four experiments perform relatively poorly, with none reaching the 60% threshold. Further analysis is provided in Section 3.2.

Overall, all four experiments exhibit strong forecasting effectiveness during this transitional weather event. Among them, CMOST and FETA achieved the highest scores across the evaluation metrics, while WMOST and WETA lagged slightly behind. In terms of the RMSE indicator under the same version of WRF, which is of primary concern regarding the grid for wind resource forecasting, the WRF_FITCH model using the WFP method (FETA) achieved the best results among the WRF models, followed by the WRF model using the ST method (WMOST), and then the unmodified WRF model (WETA).

3.2. Differential Forecasting Performance on Wind Speed Fluctuation Characteristics

The fluctuation and turning points of wind speed significantly influence wind power ramping events, which are critical to the stability of the power grid. These events are characterized by attributes such as the wind's ramp magnitude, duration, rate, and direction [15]. Drawing on those concepts, this study contrasts the wind speed fluctuation magnitude, duration, direction, and dispersion characteristics between forecasted and observed wind speeds. By employing time series analysis, it reveals the dynamic trends of wind speed fluctuations and explores the changing processes during these fluctuations.

Figure 3 displays a box plot comparing observations and forecasts at four wind farms over 72 h. Given the occurrence of extreme wind speed values that significantly deviate from the average during the research period, no outliers were omitted in the calculation of quartiles. Consequently, all data were included in the box plot representation. The plot differentiates the observed values (black) and forecasts by the CMOST scenario (red), WMOST scenario (yellow), FETA scenario (blue), and WETA scenario (green). The observed average wind speed across the stations ranged from 2.24 to 10.65 m/s. The shape of the box, spanning from the 25th to the 75th percentile, demonstrates that the middle 50% of simulation data from the WMOST and FETA experiments closely align with the actual wind speed fluctuations, whereas CMOST and WETA slightly overestimate the central range. In terms of median and mean values, the simulation results from all four experiments are relatively close but tend to be slightly overestimated.

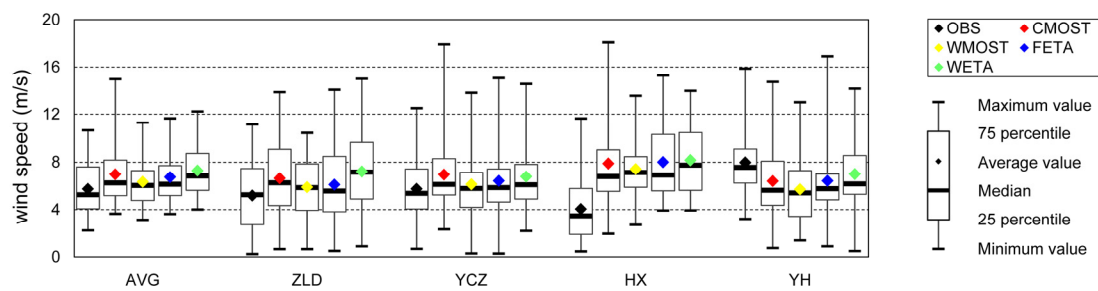


Figure 3. Box plot for observations (black) and CMOST (red), WMOST (yellow), FETA (blue), and WETA (green) scenarios' prediction over 72 h.

However, the performance differs for the maximum and minimum values, with specific wind speeds and deviation ratios detailed in Table 4, where the values with the lowest relative deviation are highlighted in bold. CMOST significantly overestimates both the average maximum and minimum values by over 40%, while WMOST, using the same ST method, performs the best, with a maximum wind speed overestimation of only 6% and a minimum wind speed overestimation of 40%. FETA shows slightly higher deviations than WMOST, while WETA's deviations are higher than both but lower than those of CMOST.

Table 4. Maximum and minimum hub-height wind speeds across four wind farms. The values with the lowest relative deviation in the same station are highlighted in bold.

INDEX	STATION	OBS	CMOST	WMOST	FETA	WETA
MIN	AVG	2.24	3.66	3.13	3.64	4.02
	ZLD	0.25	0.67	0.67	0.51	0.90
	YCZ	0.69	2.33	0.30	0.29	2.19
	HX	0.47	1.96	2.72	3.94	3.95
	YH	3.22	0.76	1.40	0.90	0.50
MIN BIAS RATIO	AVG	/	64%	40%	63%	80%
	ZLD	/	168%	169%	104%	260%
	YCZ	/	238%	−57%	−58%	217%
	HX	/	317%	479%	738%	740%
	YH	/	−76%	−57%	−72%	−84%
MAX	AVG	10.65	14.97	11.29	11.65	12.26
	ZLD	11.14	13.87	10.43	14.07	15.00
	YCZ	12.52	17.92	13.82	15.07	14.56
	HX	11.65	18.09	13.56	15.28	13.98
	YH	15.88	14.74	13.02	16.92	14.17
MAX BIAS RATIO	AVG	/	41%	6%	9%	15%
	ZLD	/	25%	−6%	26%	35%
	YCZ	/	43%	10%	20%	16%
	HX	/	55%	16%	31%	20%
	YH	/	−7%	−18%	7%	−11%

Regarding individual station performance, the box plots for ZLD align closely with the average data distribution, demonstrating relatively effective forecasting for all experiments. At the YCZ wind farm, the performance of three experiments is similar to the average data, with the exception of CMOST, which overestimates the maximum values. Significant discrepancies were noted at the HX and YH wind farms. An examination of the box sizes and antennae lengths indicates that, at the HX site, all experiments significantly overestimate the central range of wind speeds as well as the maximum and minimum values, suggesting that the current physical parameterization schemes are less suitable for this wind farm, leading to systematically higher simulated wind speeds and contributing to multiple instances of non-compliance in the verification results. Conversely, at the YH site, all models underestimate the central range of wind speeds and the minimum values.

Building on the concept of wind power ramping related to increases and decreases in wind speed [15], this study defines an upward fluctuation of wind speed as an increase and a downward fluctuation as a decrease. The magnitude of wind speed fluctuation is determined by the change in the wind speed from below the cut-in speed to its maximum, or from the maximum down to below the cut-in speed. This definition is situated within the context of the wind turbines' power change curve, which requires that the change in speed exceeds 4 m/s. Considering the variation in cut-in wind speeds across different wind turbine models, a uniform cut-in wind speed of 3 m/s is applied for this analysis.

Figure 4 illustrates the time series of the RMSE between the observed average wind speeds and forecasted wind speeds for all wind farms over a 72 h period. The data identify two significant fluctuation episodes. Specifically, from 071002 to 071016 UTC, a downward fluctuation, reaching a maximum wind speed of 10.6 m/s and a fluctuation magnitude of 7.6 m/s over 14 h, termed downward fluctuation 1, was observed. This was followed by an upward fluctuation from 071107 to 071122 UTC, achieving a maximum wind speed of 9.2 m/s and a fluctuation magnitude of 6.2 m/s over 15 h, named upward fluctuation 1. Immediately thereafter, from 071122 to 071204 UTC, a second downward fluctuation was observed with the same maximum wind speed and fluctuation magnitude as the preceding upward fluctuation 1, which lasted only 6 h and was referred to as downward fluctuation 2. The magnitude of these fluctuations was approximately 7.0 m/s, yet the rate of change during downward fluctuation 2 was significantly faster than that during the previous fluctuations, demonstrating a sudden drop in wind speed.

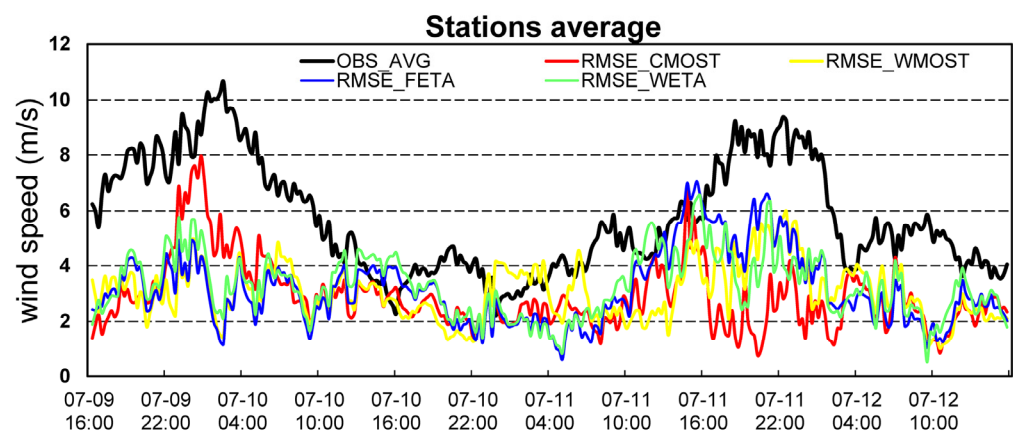


Figure 4. Time series of RMSE of CMOST (red), WMOST (yellow), FETA (blue), and WETA (green) scenarios among all stations over 72 h (070916 to 071216 UTC), and the corresponding average observation (black).

The RMSE series in Figure 4 indicates that all four forecasting methods maintained good stability throughout the entire 72 h prediction window, with the forecast errors not being significantly affected by fluctuation phases or the duration of forecasts. The RMSE error values corresponded to periods of high wind speed. Among the scenarios, FETA and WETA demonstrated relative stability, whereas CMOST and WMOST, both using the ST method, had periods where their RMSE values were noticeably higher than those of the others, such as during the period from 070922 to 071004 UTC. CMOST exhibited the largest RMSE fluctuations, with values reaching up to 8 m/s at certain times, while the other three experiments remained more stable, with RMSE values staying below 7 m/s.

Figure 5 presents a time series comparison of observed and forecasted wind speeds at four wind farms, as well as their overall average. The direct comparison of curves reveals synchronous changes and wind speed transitions that were not captured by earlier statistical indicators. For the station average, at 071100 UTC, when the observed wind speed was at a low of 2 m/s, WMOST provided a significantly higher forecast of 7 m/s. This mismatch between forecasted and observed wind speed extremes is even more pronounced in individual station simulations. Although the CMOST method demonstrated smaller

errors and higher correlation coefficients in statistical indicators, it failed to match the observed timing of minimum wind speeds at the YCZ, HX, and YH wind farms. Similarly, FETA and WETA exhibited this issue at the YCZ wind farm.

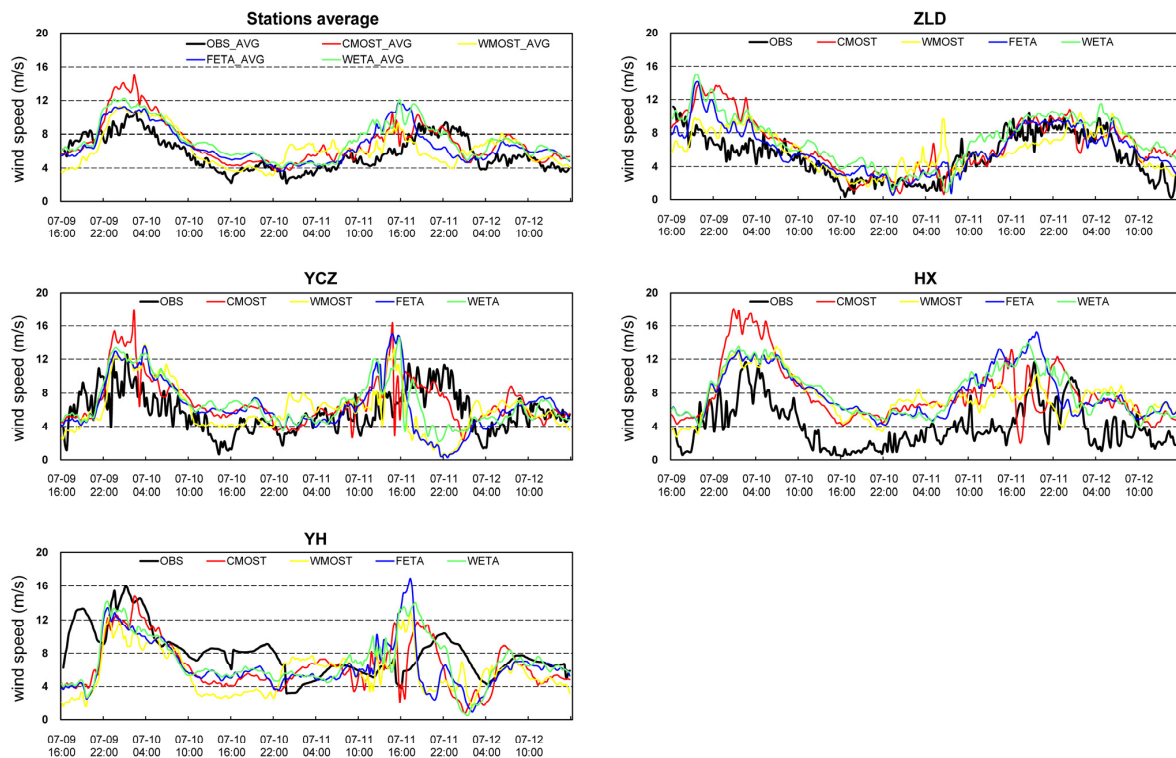


Figure 5. Time series of observed and forecasted wind speeds at each station over 72 h (07091600 to 071216 UTC), black line for observation, red for CMOST, yellow for WMOST, blue for FETA, and green for WETA.

Overall, the four models effectively simulate the general fluctuation trends of hub-height wind speeds in the BTH region. However, the CMOST and WMOST scenarios, which used the ST method, exhibit greater error fluctuations, while FETA and WETA are more stable. All model scenarios show issues with mismatches between the forecasted and observed timings of extreme wind speed values during certain local periods. Therefore, accurately forecasting the turning points of wind speed fluctuations in specific local periods remains a significant challenge for NWP.

3.3. Spatial Variability in Wind Power Density Distribution

Wind power density (WPD) represents the flow of wind energy passing through a specific area perpendicular to the airflow within a specified time interval. In this study, we employ the average wind power density to estimate the WPD at grid points at hub-height across the simulation area [13]. The model layer is set at a hub-height of 80 m. Instead of using the standard air density, we adopt the model layer air density. Here, $\bar{v}$ denotes the average wind speed at the model layer over a 72 h period.

$$P_d = \frac{1}{2} \rho \bar{v}^3 \quad (18)$$

This study assesses the influence of two methodologies on the spatial distribution patterns of the WPD within the core simulation zone. We calculate the mean spatial distribution of the WPD over a 72 h timeframe and analyze the variance, and the results are shown in Figure 6. Figure 6a–d demonstrates that the average distributions of all scenarios exhibit a distinct east-high–west-low gradient in the WPD: in the western region, the WPD

ranges from 0 to 20 W/m²; in the central region, it varies from 200 to 400 W/m²; and in the eastern region, it exceeds 400 W/m². In particular, the western region encompasses the Taihang mountains and is part of the Jing-Jin-Ji area, while the central area consists of plains (as shown in Figure 1b). The WRF model employs an eta coordinate system, which aligns vertically with the terrain and contributes to this gradient [45]. This alignment leads to a decrease in the air density at higher altitudes, resulting in a diminished WPD in the north-west mountainous regions. However, the eastern Bohai Sea area, characterized by minimal surface roughness and reduced obstruction effects compared to land, experiences enhanced wind speeds over the sea [46,47]. These conditions notably elevate the WPD in the eastern region, illustrating the complex interplay between geography and model predictions.

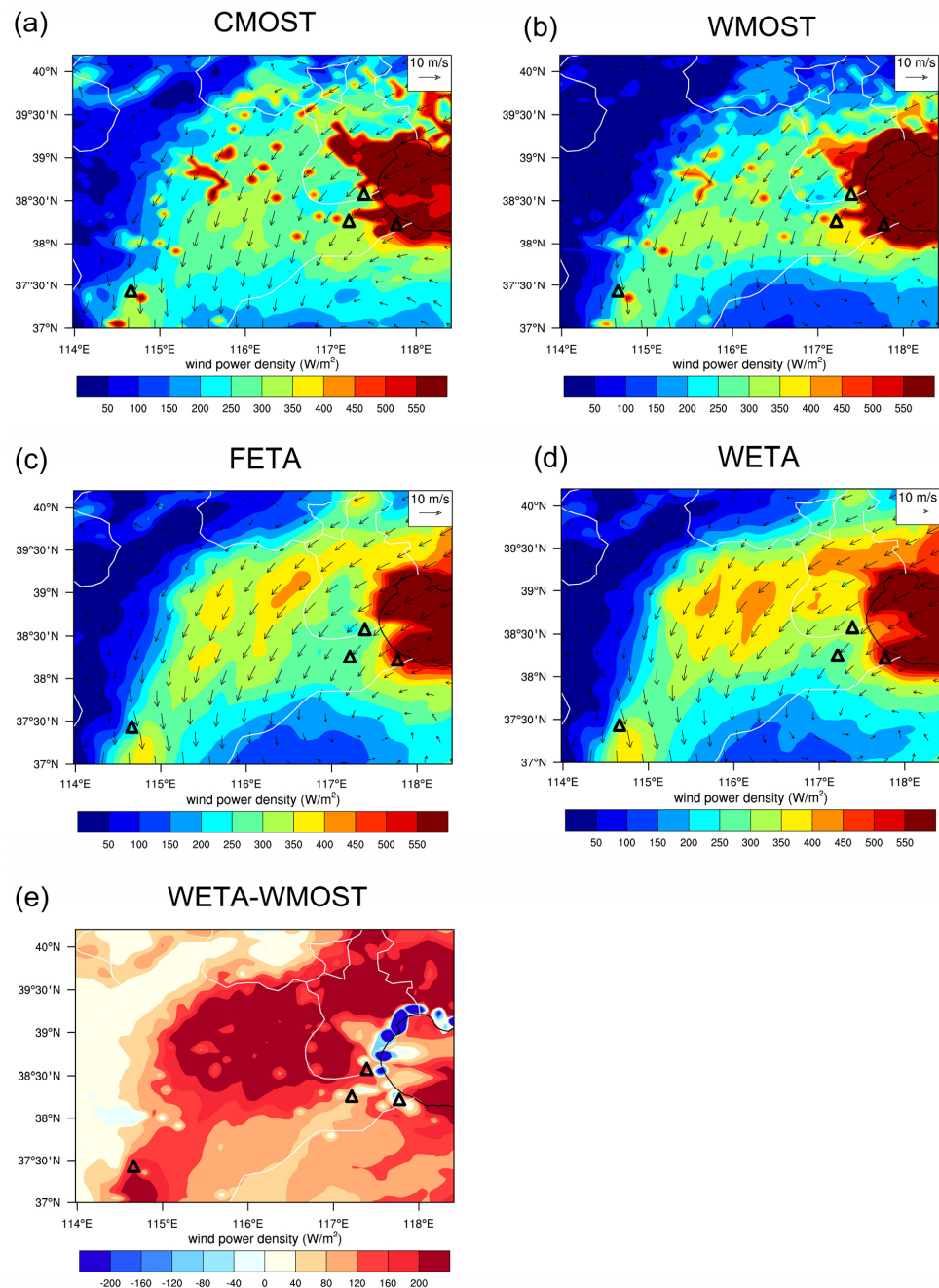


Figure 6. Hub-height wind power density forecast average distribution over 72 h: for the CMOST scenario (a), the WMOST scenario (b), the FETA scenario (c), the WETA scenario (d), and the difference of WETA minus WMOST (e). Black triangles represent the locations of wind farms, arrows represent wind vector.

In terms of the overall magnitude of WPD forecasts across the region, CMOST and WMOST are slightly worse than FETA and WETA. In most land areas, the WPD values calculated by the former two models range from 0 to 400 W/m², while those calculated by the latter two range from 0 to 450 W/m². The investigation reveals a more pronounced discrepancy between WETA and WMOST in the central and northeastern plain region, where the WETA is significantly greater than the WMOST, exceeding it by more than 120 W/m², as depicted in Figure 6e. Conversely, at the land–sea boundary, the WFP is more than 120 W/m² lower than ST.

3.4. Analysis of the Influence of the PBL on Hub-Height Wind Speed Forecasts

The discrepancies in the spatial distribution of hub-height wind speeds between the WETA and WMOST primarily stem from their differing responses to the boundary layer dynamics, which are accentuated by abrupt changes in the underlying surface from sea to land [4,48]. The WETA method relies on the boundary-layer scheme, while the WMOST method relies predominantly on a near-surface-layer scheme. Both schemes are essential physical processes in numerical weather prediction (NWP) and directly affect the forecasting of wind speeds at wind turbine hub-heights. The WMOST method takes into account atmospheric stability and surface roughness, but it mainly depends on near-surface 10 m wind speeds, resulting in a less-accurate representation of higher-level wind speeds.

Moreover, this interaction becomes critical at an elevation of 85 m, where the near-surface-layer transitions into the bottom part of the mixed-boundary layer, thereby highlighting the boundary layer's contributions to the wind field. The WETA method integrates complete advection and turbulent transport, fully accounting for the exchange processes and feedback between the two layers. In contrast, the WMOST method primarily extrapolates the near-surface wind speed using a similar theory but does not adequately address disturbances at higher wind speeds, thus neglecting essential aspects of boundary layer physical processes. Typically, turbulence directly proportional to the wind speed tends to amplify disturbances during wind speed fluctuation periods. Given that the maximum instantaneous wind speed reached 16 m/s, it can be inferred that increased turbulence significantly affects hub-height wind speeds, leading to the notable differences observed between the two methods in the central plains area [18,48].

Therefore, wind resource assessments at turbine hub-heights should extend beyond surface wind speeds and profiles to include a thorough consideration of the boundary layer's influence. Future improvements could involve refining surface roughness features by incorporating an implicit wind turbine parameterization scheme and updating the original land-use types to better reflect wind farms' equivalent surface roughness. Such enhancements would improve modelling and better characterize the impact of wind farms on the atmospheric flow field [4,49]. Given the strong locality and scarcity of vertical wind speed observations at wind farm hubs, the limited availability of observation sites significantly impedes most research into wind speeds at hub-heights. Therefore, collecting hub-height observation data across various terrains is essential for advancing wind energy forecast research and its validation.

4. Conclusions

This study conducted a comprehensive comparison of four scenario experiments designed with a 9 km resolution in the BTH region of China, using three numerical weather prediction (NWP) models: CMA_WSP, WRF, and WRF_FITCH. The experiments applied different hub-height wind speed calculation methods, including the Monin–Obukhov similarity theory (ST) method and the wind farm parameterization (WFP) method, as well as other fundamental approaches. It focuses on the performance of wind speed and wind power density (WPD) at hub-height, which is especially critical during summer when electricity demand peaks and the wind speed variability is substantial. Data from four Hebei wind farms were analyzed to assess the forecast accuracy over a 72 h period, highlighting the transitional weather characterized by significant fluctuations, with maximum average

wind speeds exceeding 10 m/s. This analysis provides valuable insights into wind speed behaviour under transitional weather patterns.

Verification was conducted using standard numerical model and power grid evaluation criteria including the RMSE, correlation coefficient (R), accuracy rate (AR), and qualification rate (QR). All four experiments demonstrated strong forecasting effectiveness during the transitional weather event. Among the models, CMOST and FETA achieved the highest scores across the evaluation metrics, indicating superior performance. Specifically within the WRF series models, the WRF_FITCH model using the WFP method (FETA) achieved the best results. FETA exhibited the lowest RMSE, followed by WMOST, which uses the ST method, and then the unmodified WRF model (WETA). Despite all WRF models producing stable and accurate forecasts, FETA consistently outperformed the others, particularly in minimizing the RMSE and achieving better correlation, accuracy, and qualification rates.

Furthermore, the four models showed consistency with observed hub-height wind speed data and effectively captured the general fluctuation trends across the entire BTH region. However, CMOST and WMOST, using the ST method, showed greater error fluctuations, while FETA and WETA were more stable. All models struggled with accurately forecasting the timing of extreme wind speeds during certain local periods, highlighting a persistent challenge in NWP models. This study also explored the performance of wind energy forecasting models at hub-height, particularly focusing on their ability in calculating the spatial distribution of the average WPD. The differences observed between WETA and WMOST, especially in the central and northeastern regions, were attributed to the dynamics of the boundary layer. The WETA scenario integrates complete advection and turbulent transport, enhancing the interaction between the boundary and near-surface layers, whereas the WMOST method relies on extrapolating the near-surface wind speed.

This study underscores the critical importance of considering boundary layer dynamics to refine wind speed forecasts and enhance wind energy resource estimations at higher hub-heights. This integrated approach will enhance the precision of mesoscale forecasting methods, support wind farm operation and maintenance, and optimize wind energy resource utilization. The conclusions drawn from this study are specific to transitional weather conditions characterized by wind speed fluctuations and are subject to certain uncertainties. Future research should expand observational data collection and undertake broader numerical forecasting experiments to optimize and validate wind speed and energy forecasts at hub-heights across varied weather conditions.

Author Contributions: J.M.: Investigation, Methodology, Software, Validation, Writing-Original Draft; Y.S.: Resources, Conceptualization, Writing-Review & Editing; B.Y.: Software, Writing-Review & Editing; M.L.: Software; C.D.: Writing-Review; B.J.: Software; D.Y.: Software; D.W.: Investigation. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by Research Project of the Tibet Autonomous Region Association for Science and Technology (Large-Scale Development of Ultra-High Altitude Wind Power in the Tibet Region), the Innovation Foundation of Public Meteorological Service Center of China Meteorological Administration (grant number Y2023007), the Key Innovation Team for Energy and Meteorology of China Meteorological Administration, the Natural Science Foundation (grant number 8234069), and the Research Project of Shaanxi Provincial Department of Science and Technology (grant number 2023-YBSF-235).

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The data presented in this study are available on request from the corresponding author. The data are not publicly available due to privacy.

Conflicts of Interest: The authors declare that they have no conflicts of interest.

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